PROPERNESS DESCENDS ALONG AN ARBITRARY FIELD EXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter text immediately before the lemma notes that it is a special case of the descent lemma for properness; that remark lies outside the counted span and enters only as context.

Editorial note. The counted spans carry 11 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-proper-after-field-extension is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Properness is stable under base change and composition. Morphisms, Lemmas morphisms-lemma-base-change-proper and morphisms-lemma-composition-proper, invoked at proof lines 2878-2884. Supplies that X_{k'} is proper over k' and that the composite X_{k'} to Spec(k) is proper.
- Properness descends through a filtered colimit of rings. Limits, Lemma limits-lemma-eventually-proper, invoked at proof lines 2871-2872. Supplies the index i for which X_i over Spec(A_i) is already proper.
- Hilbert Nullstellensatz: the residue field at a maximal ideal of a finitely generated k-algebra is a finite extension of k. Algebra, Theorem algebra-theorem-nullstellensatz, invoked at proof lines 2875-2877. Supplies the finite extension k' of k used for the final descent.
- Image of a proper morphism and separatedness of the source. Morphisms, Lemmas morphisms-lemma-image-proper-is-proper and morphisms-lemma-image-universally-closed-separated, invoked at proof lines 2885-2889. Supplies separatedness and properness of X over k from those of X_{k'} to X.

Source reference. Original theorem 00FV: sources/stacks-chow/tags.txt, source lines 585–585. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 081F: sources/stacks-chow/tags.txt, source lines 10215–10215. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Source reference. Original definition 01W1: sources/stacks-chow/tags.txt, source lines 2466–2466. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01W4: sources/stacks-chow/tags.txt, source lines 2469–2469. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01W3: sources/stacks-chow/tags.txt, source lines 2468–2468. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01WN: sources/stacks-chow/tags.txt, source lines 2488–2488. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 03GN: sources/stacks-chow/tags.txt, source lines 4534–4534. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 09MQ: sources/stacks-chow/tags.txt, source lines 12188–12188. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 01PS: sources/stacks-chow/tags.txt, source lines 2241–2241. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01KO: sources/stacks-chow/tags.txt, source lines 2057–2057. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be a field. Let X be a scheme over k. If X_K is proper over K for some field extension K/k, then X is proper over k.*

Proof. Let K/k be a field extension such that X_K is proper over K. Recall that this implies X_K is separated and quasi-compact (Morphisms, Definition 01W1). The morphism $X_K \to X$ is surjective. Hence X is quasi-compact as the image of a quasi-compact scheme (Properties, Definition 01PS). Since X_K is separated we see that X is quasi-separated: If $U, V \subset X$ are affine open, then $(U \cap V)_K = U_K \cap V_K$ is quasi-compact and $(U \cap V)_K \to U \cap V$ is surjective. Thus Schemes, Lemma 01KO applies.

Write $K = \operatorname{colim} A_i$ as the colimit of the subalgebras of K which are of finite type over k. Denote $X_i = X \times_{\operatorname{Spec}(k)} \operatorname{Spec}(A_i)$. By Limits, Lemma 081F there exists an i such that $X_i \to \operatorname{Spec}(A_i)$ is proper. Here we use that X is quasi-compact and quasi-separated as just shown. Choose a maximal ideal $\mathfrak{m} \subset A_i$. By the Hilbert Nullstellensatz (Algebra, Theorem 00FV) the residue field $k' = A_i/\mathfrak{m}$ is a finite extension of k. The base change $X_{k'} \to \operatorname{Spec}(k')$ is proper (Morphisms, Lemma 01W4). Since k'/k is finite both $X_{k'} \to X$ and the composition $X_{k'} \to \operatorname{Spec}(k)$ are

proper as well (Morphisms, Lemmas 01WN, 01W4, and 01W3). The first implies that X is separated over k as $X_{k'}$ is separated (Morphisms, Lemma 09MQ). The second implies that $X \to \mathrm{Spec}(k)$ is proper by Morphisms, Lemma 03GN. $\square$